Objects and editing

A flow subtracts from one named account and adds to another. The application uses Python dictionaries and one SQLite repository. These are the actual objects returned by Repository.get_account() and get_flow().

```
{
"name": "AmEx",
"opening_cents": 0,
"opening_at": "2026-10-01 00:00:00"
}

{
"flow_id": "11111111-1111-4111-8111-111111111111",
"date": "2026-10-02 12:30:00",
"from_account": "AmEx", "to_account": "Joe's Diner",
"amount_cents": 1200, "status": "Complete",
"name": "Friday lunch", "notes": "Burger and fries",
"category": "Dining", "tags": ["Road trip"]
}
```

Representation	Meaning
Account name	Its identity. Starting balance is integer cents, negative for debt, or null when unknown. New endpoint accounts start unknown.
Amount / flow ID	Input .50 means 50 cents; stored 1200 means $12.00. Exact integer arithmetic. UUID preserves identity during edits/imports. Repeat creates a new UUID.
Date	One Date/Time entry, in 24-hour format. Leading zeros optional. Missing month/day = 1; missing hour/minute/second = 0. Example: 2026-10-9 7:5 becomes 2026-10-09 07:05:00.
Category / tags	One category and a list of tag names. Shared tags can connect split purchases. Nine flow fields plus ID; no scenario field.

The GUI edits these objects

Date/Time, From, To, Amount, Status, Name, Notes, Category, Tags. Defaults: today, Complete and Uncategorized. Accounts has Name, Starting Balance and Starting Date/Time. No separate Time control.

Existing records save on Enter, Tab or leaving a field. Ctrl+S finishes existing edits and leaves new drafts untouched. Notes: Enter adds a line; Ctrl+Enter or leaving/closing saves. Invalid edits stay unsaved with a silent error; Escape restores the saved value.

New Flow clears; Add Flow inserts; Repeat copies every value. Empty ledger space deselects and clears, saving valid edits and discarding invalid/unadded drafts. Row/page changes discard new drafts; undo restores them. Drag table titles to move columns; cell editing and Tab follow their visible order.

Repository-generated objects and rows: examples/record_objects.json and examples/review.sql. Launching and controls: README.md. Source walkthrough: docs/model_review.md. Forecast collections and simulation controls are deferred.

Storage and calculations

Named keys are name.strip().casefold(): AmEx becomes amex. They provide case-insensitive matching and foreign-key references, not separate generated identities. Display spelling stays in name.

Table	Every stored column
accounts	account_key, name, opening_cents, opening_at
flows	flow_id, date, from_key, to_key, amount_cents, status, name, notes, category_key
categories	category_key, name
tags	tag_key, name
flow_tags	flow_id, tag_key, position
audit	audit_id, changed_at, entity, entity_key, action, before_json, after_json

Category/tag tables retain choices; flow_tags preserves assignments and order. audit stores before/after values. Account lists show only names used in saved From/To fields, including Void flows. Unreferenced starting balances remain stored for reuse/undo. Schema: flowledger/schema.sql.

Save and calculate

GUI draft -> Repository.save_flow() -> validate -> write flow, choices, tags and audit atomically. Failed saves roll back. Ctrl+Z / Ctrl+Y undo records, drafts, filters, searches, dates, sorting, graph controls and column changes in action order. history.py captures changed rows; view_history.py keeps UI snapshots in memory. Record undo appends audit; closing/restoring clears history. No new stored fields.

An account receives +amount when To and -amount when From. Confirmed includes Complete; Including Pending adds Pending; Including Planned also adds Planned. Void contributes zero. Settle Pending by editing that record; a refund is a new reverse flow.

Each account has one confirmed starting balance immediately before flows at its timestamp. Earlier Complete flows are already reflected; eligible earlier Pending/Planned flows enter estimates once. Unknown balances remain unknown. Dates are points in local time, compared at second precision; date-only means midnight.

Graphs compare or sum explicitly selected accounts. A transfer cancels within a sum containing both endpoints. Selecting every endpoint conserves the total; this is not automatically net worth. Category totals count each matching non-Void flow once, including transfers.

Upgrade and scope

Schema 1-3 upgrades first save a full .pre-v4-timestamp.bak. Old Baseline flows stay active with their IDs; accounts and history remain. Old hypothetical collections remain in the backup and .bak.forecasts.json recovery file, outside the active ledger. Time flags become date precision, then disappear.

Import/export: nine fields plus ID. Old scenario/time-flag imports are rejected; upgrade the database and export again. Use Backup for a complete snapshot. Scope: USD cents, one starting balance per account, local timestamps, one computer. Executed checks: docs/test_results.txt and docs/interaction_check.txt.